

Week 4 Weekly Report

Week 4 established the baseline evaluation for the chooser pricing project without breaking the data and model chain built in Weeks 1 through 3. The work separates computational validation from market validation and does not compare an option value with the JPM stock price.

1 Work Completed

- Loaded the Week 2 processed JPM, VIX, and Treasury dataset and called the Week 3 simple chooser pricing model through a regression-tested adapter.
- Refactored the chooser functions into a vectorized module while preserving Week 3 outputs exactly.
- Generated 1,740 historical chooser baseline prices using contemporaneous JPM spot, Treasury rate, and trailing 20-day volatility.
- Built a forward-volatility proxy benchmark and evaluated it by VIX regime.
- Performed a separate listed JPM vanilla option sanity check with delayed Cboe quotes; this check does not claim to validate chooser transaction prices.
- Measured runtime, throughput, memory use, and asymptotic complexity.

2 Deliverables

- Week4_Baseline_Evaluation.ipynb
- bsm_chooser.py
- week3_adapter.py
- benchmark_config.json
- model_results and figures
- Model Validation Report
- Performance Benchmark Documentation

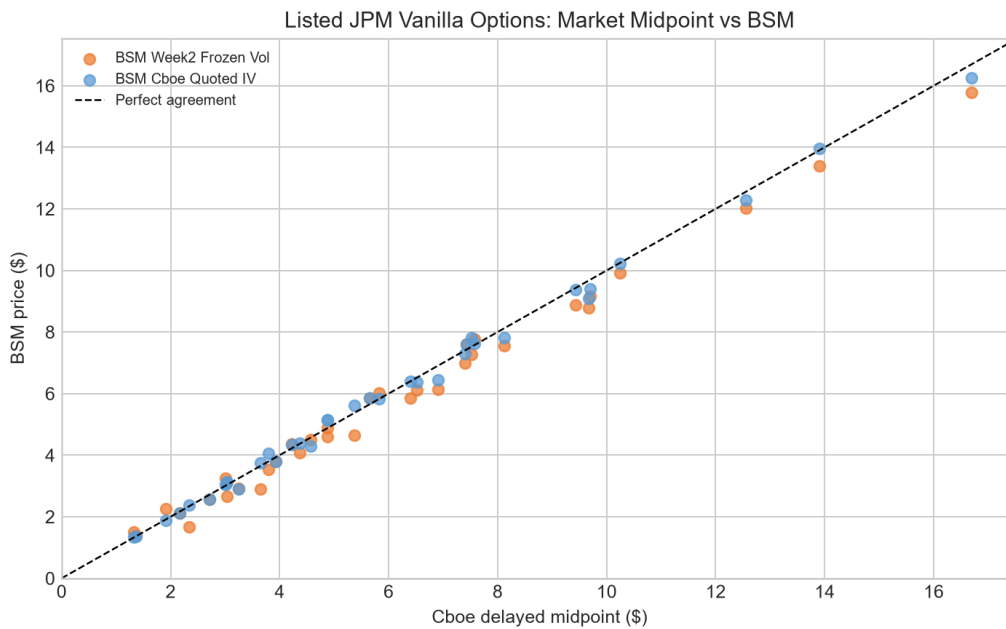
3 Progress Summary

All planned Week 4 baseline tasks that can be completed with defensible data are complete. The Week 3 pricing logic matches the optimized implementation with zero numerical difference. The computational benchmark and listed-option market sanity check now provide a documented reference for the Week 5 and Week 6 machine-learning models.

4 Experimental Results

Validation	N	MAE	RMSE	R ²
Historical chooser proxy	1,720	MAE 3.7757	RMSE 8.3567	R ² 0.8242
Cboe vanilla with Week 2 volatility	36	MAE 0.3635	RMSE 0.4401	R ² 0.9845
Cboe vanilla with quoted IV	36	MAE 0.1658	RMSE 0.2201	R ² 0.9961

Figure 1 Listed JPM option market sanity check



5 Issues Encountered

- Public historical transaction prices for a customized JPM chooser option were not available. The report therefore labels forward-volatility comparisons as proxy validation.
- The project brief mentions CME transaction prices even though the underlying is JPM stock. The evaluation remains on the JPM data chain and uses delayed listed JPM vanilla quotes only as a separate BSM market sanity check.
- The fixed \$150 strike becomes deep in the money as JPM rises, reducing volatility sensitivity in later observations.

6 Plan for Week 5

- Design two leakage-safe machine-learning approaches.
- Create a chronological 70/15/15 split.
- Preserve the Week 4 BSM price as the benchmark and structural anchor.